Inorganic and Materials Cumulative Exam
November 1, 2024

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. (a) (10pts) Describe the electronic structure of HF using a molecular orbital diagram. (F 1s orbital potential = -40.2eV, F 2s orbital potential = -18.6eV, H 1s orbital potential = -13.6eV).
(b) (5pts) Sketch the electron density of the HOMO and LUMO levels of the HF molecule.
(b) (5pts) Can you used a molecular orbital diagram to describe the compound LiF? Why or why not? (Li 2s orbital potential = -5.39eV).

2. (a) (15pts) Define linkage isomerism and provide an example of a ligand that can participate in it. Draw diagrams to support your answer.
(b) (5pts) Describe the role that the solvent can play in the bonding of a metal-ligand complex that exhibits linkage isomerism.

3. The reduction potential of decamethylcobaltocene $(Co(C_5Me_5)_2)^{+/0}$ is nearly 600 mV more negative than cobaltocene $(Co(C_5H_5)_2)^{+/0}$. Explain this observation.

4. (a) (6 pts) Draw the molecular orbital diagram of C_2.
(b) (6 pts) Calculate the bond order of C_2, C_2^+, and C_2^{2+}.
(c) (8 pts) Of the three dicarbon molecules in (b), which has the shortest bond length? Justify your answer – an answer with no justification gets no credit.

5. (a) (4 pts) What is the point group of NH_3? (hint: it's not planar)
(b) (8pts) Using the appropriate character table (included), how many IR-active modes does NH_3 have?
(b) (8pts) Using the appropriate character table (included), how many Raman-active modes does NH_3 have?

6. Derive Bragg's law for the interference pattern of x-rays scattering from two parallel planes. This is a simple basis for understanding x-ray diffraction, for example.

7. (a) (12pts) In the context of mixed-metal solid solutions, what does it mean for the components to be immiscible?
(b) (8pts) Explain how two metals may be immiscible in a bulk material, yet miscible in a nanoparticle formulation.

8. (a) What is an exciton?
(b) Excitons can exist in semiconductors and in polymers, but their lifetimes in polymers are orders of magnitude longer. Why is this?

9. How do you expect the Cu 2p binding energy (or XPS peak energy, if you prefer to discuss it that way) to shift for the series of copper oxidation states – Cu (metal), Cu_2O (s), CuO(s)? Explain your answer using specific chemical and physical detail.

10. A simple derivation of one-dimensional band theory involves building exceedingly longer chains of identical potential wells. In the figure below, (a) shows the ground state, 's-like' electronic wavefunction of a finite square well potential and its ground state energy E_0. Using this graphical result,

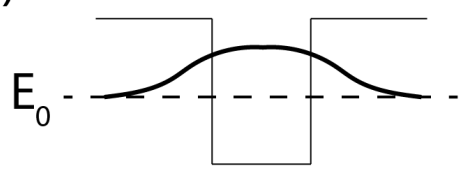
(a) (6 pts) Draw the unique 'linear combinations of atomic orbitals' that represents the ground state for the four finite-square well chain (b).

(b) (6 pts) Estimate the energy of the ground state you described in (a) with respect to E_0 – is it higher or lower than E_0? What is your rationale?

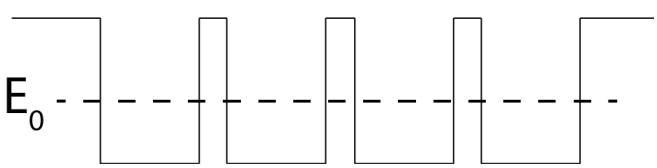
(c) (8 pts) Draw three other <u>orthogonal</u> linear combinations of the four finite square well chain and rank them in energy from lowest to highest.

(Figure for question #10)

(a)



(b)



Some useful character tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v'(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	-1	R_z	
E	2	-1	0	$(x, y), (R_x, R_y)$	$(x^2 - y^2, xy), (xz, yz)$

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		$x^2 - y^2$
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	$(x, y), (R_x, R_y)$	(xz, yz)

D_2	E	$C_2(z)$	$C_2(y)$	$C_2(x)$		
A	1	1	1	1		x^2, y^2, z^2
B_1	1	1	-1	-1	z, R_z	xy
B_2	1	-1	1	-1	y, R_y	xz
B_3	1	-1	-1	1	x, R_x	yz

D_3	E	$2C_3$	$3C_2$		
A_1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	-1	z, R_z	
E	2	-1	0	$(x, y), (R_x, R_y)$	$(x^2 - y^2, xy), (xz, yz)$

D_{2d}	E	$2S_4$	C_2	$2C_2'$	$2\sigma_d$		
A_1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		$x^2 - y^2$
B_2	1	-1	1	-1	1	z	xy
E	2	0	-2	0	0	$(x, y), (R_x, R_y)$	(xz, yz)

D_{3d}	E	$2C_3$	$3C_2$	i	$2S_6$	$3\sigma_d$		
A_{1g}	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_{2g}	1	1	-1	1	1	-1	R_z	
E_g	2	-1	0	2	-1	0	(R_x, R_y)	$(x^2 - y^2, xy)$ (xz, yz)
A_{1u}	1	1	1	-1	-1	-1		
A_{2u}	1	1	-1	-1	-1	1	z	
E_u	2	-1	0	-2	1	0	(x, y)	

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	−1	−1	1	1	−1	−1	R_z	xy
B_{2g}	1	−1	1	−1	1	−1	1	−1	R_y	xz
B_{3g}	1	−1	−1	1	1	−1	−1	1	R_x	yz
A_u	1	1	1	1	−1	−1	−1	−1		
B_{1u}	1	1	−1	−1	−1	−1	1	1	z	
B_{2u}	1	−1	1	−1	−1	1	−1	1	y	
B_{3u}	1	−1	−1	1	−1	1	1	−1	x	

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2'	1	1	−1	1	1	−1	R_z	
E'	2	−1	0	2	−1	0	(x, y)	$(x^2 - y^2, xy)$
A_1''	1	1	1	−1	−1	−1		
A_2''	1	1	−1	−1	−1	1	z	
E''	2	−1	0	−2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_{2g}	1	1	1	−1	−1	1	1	1	−1	−1	R_z	
B_{1g}	1	−1	1	1	−1	1	−1	1	1	−1		$x^2 - y^2$
B_{2g}	1	−1	1	−1	1	1	−1	1	−1	1		xy
E_g	2	0	−2	0	0	2	0	−2	0	0	(R_x, R_y)	(xz, yz)
A_{1u}	1	1	1	1	1	−1	−1	−1	−1	−1		
A_{2u}	1	1	1	−1	−1	−1	−1	−1	1	1	z	
B_{1u}	1	−1	1	1	−1	−1	1	−1	−1	1		
B_{2u}	1	−1	1	−1	1	−1	1	−1	1	−1		
E_u	2	0	−2	0	0	−2	0	2	0	0	(x, y)	